

Can Language Models Forecast Their Own Belief Updates?

A Two-Context Test of Metacognitive Forecast Revision

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Abstract

Probabilistic forecasts are useful only insofar as they can be revised under new evidence. Recent work studies whether language models can forecast future events, whether they update beliefs in a Bayes-like manner, and whether they can predict aspects of their own future behavior. We connect these threads with a direct metacognitive test: can a language model predict how its own forecast will change after seeing evidence? For each forecasting question X and neutral evidence packet Y , we first elicit the model’s current probability $\pi = P(X)$. In the same context, before allowing an ordinary update, we ask the model to predict what its own posterior would be after receiving Y . In a fresh context, we then actually provide Y and elicit the realized posterior. We compare predicted and realized updates in log-odds space. On a 200-question ForecastBench-derived validation set using GPT-OSS-120B, the model’s predicted and realized log-odds updates have Spearman correlation 0.835 and reduce mean absolute log-odds error by 58.0% relative to a no-update baseline. After removing exact 0/1 parsed posteriors, Pearson correlation rises from 0.664 to 0.899 and Spearman correlation remains high at 0.855. Error analysis shows that failures concentrate in a small number of extreme posterior shifts, while ordinary-sized updates remain strongly predictable. These results suggest that language models exhibit measurable self-forecasting of their own evidence-conditioned forecast revisions, without implying Bayes optimality or privileged access to internal beliefs.

1 Introduction

Forecasting systems should not merely issue probabilities; they should also revise them coherently when new information arrives. Large language models (LLMs) are increasingly evaluated as forecasters on real-world questions [1, 2]. Separately, a growing literature asks whether LLMs can evaluate their own knowledge or anticipate their own behavior [3, 4, 5], and another asks whether LLMs can perform probabilistic or Bayesian-style belief updating [7]. What is missing is the direct bridge: can a model predict its own future belief update before the evidence is actually shown in an ordinary forecasting context?

We propose a simple two-context protocol. Given a binary question X , first elicit the model’s current forecast π . Then, in the same context, present a counterfactual description of evidence Y and ask the model to predict the posterior it would give after incorporating Y . Finally, in a fresh context, provide the same evidence Y and elicit the model’s actual posterior. The central object is not the posterior itself, but the update:

$$\Delta_{\text{pred}} = \text{logit}(\pi'_{\text{pred}}) - \text{logit}(\pi), \quad (1)$$

$$\Delta_{\text{act}} = \text{logit}(\pi'_{\text{act}}) - \text{logit}(\pi). \quad (2)$$

If models have metacognitive access to their own forecast revision behavior, then Δ_{pred} should track Δ_{act} across many (X, Y) pairs.

This paper reports an initial, deliberately narrow result: neutral evidence packets only, one model, and 200 validation questions. The result is nonetheless sharp. Predicted and actual updates are strongly rank-correlated, substantially closer than a no-update baseline, and robust after excluding exact boundary probabilities. The main failure mode is also interpretable: a small number of extreme actual shifts dominate RMSE and depress all-row Pearson correlation.

2 Protocol

Each example consists of a binary forecasting question X , a prior phase, and a neutral evidence packet Y dated before the forecast date. The prior phase includes the question, source background, resolution criteria, forecast date, resolution date, and a reference-class base rate. The base rate is shown as context, but in the exact protocol it is not used as π ; π is elicited from the model.

Step 1: elicit the current forecast. The model receives only the prior phase and is asked to output a probability on the first line in the form **Probability: <number between 0 and 1>**. This probability is π .

Step 2: predict the future update. In the same conversation, the model is told that it is about to receive evidence Y in a separate ordinary forecast. Before making that ordinary forecast, it is asked to predict what its own posterior forecast would become after incorporating Y . This yields π'_{pred} .

Step 3: realize the update in a fresh context. In a new context, the model receives the same prior phase, acknowledges that it will use the prior phase as its starting point, then receives Y as the update phase and outputs a posterior. This yields π'_{act} .

Metric. We compare log-odds updates from the same elicited prior. To avoid infinite values, probabilities are clipped to $[10^{-6}, 1 - 10^{-6}]$ before applying $\text{logit}(p) = \log(p/(1 - p))$. We report Pearson and Spearman correlations between Δ_{pred} and Δ_{act} , mean absolute error (MAE), RMSE, posterior probability gap, and improvement over a no-update baseline ($\Delta_{\text{pred}} = 0$).

3 Data and Model

We evaluate on 200 validation questions from the repository’s ForecastBench-derived dataset. The evidence packets are the neutral variants of the generated prior/update dataset; stronger weak-yes, strong-yes, weak-no, and strong-no framings are reserved for future experiments. We run GPT-OSS-120B via Tinker sampling at temperature 0, top- $p = 1$, and a 512-token generation cap. The run produced 199 parseable prior forecasts and 189 fully parseable prior/predicted/actual triples, giving pair coverage 94.5%.

4 Results

Table 1 gives the main result. Across all parseable pairs, predicted and actual log-odds shifts have Spearman correlation 0.835 and Pearson correlation 0.664. The model reduces log-odds MAE by 58.0% relative to predicting no update. Removing exact 0/1 posteriors leaves nearly the same number of examples but changes the magnitude-sensitive metrics substantially: Pearson rises to

Metric	All rows	Excl. exact 0/1	$ \Delta_{\text{act}} < 5$
Parseable pairs / rows	189 / 200	188 / 200	186 / 200
Pearson corr.	0.664	0.899	0.876
Spearman corr.	0.835	0.855	0.850
Log-odds MAE	0.440	0.372	0.367
Log-odds RMSE	1.080	0.503	0.491
Posterior RMSE	0.044	0.037	0.037
No-update MAE	1.048	0.987	0.942
MAE improvement vs no-update	58.0%	62.3%	61.0%

Table 1: Main exact-protocol results. The model substantially predicts its own future update. Exact boundary probabilities primarily affect Pearson and RMSE through logit clipping, while rank correlation remains high.

$ \Delta_{\text{act}} $ bin	n	Pearson	Spearman	MAE	RMSE	Post. MAE
< 1	116	0.805	0.780	0.308	0.428	0.025
$[1, 3)$	69	0.893	0.557	0.463	0.578	0.015
$[3, 5)$	1	–	–	0.693	0.693	0.00005
≥ 5	3	0.443	0.500	4.934	7.653	0.127

Table 2: Performance conditioned on actual update extremity. Most examples fall below $|\Delta_{\text{act}}| < 3$, where linear correlation and error remain strong. The extreme bin is tiny and contains the exact-zero outlier.

0.899 and RMSE falls to 0.503. This confirms that a small number of boundary outputs dominate all-row squared error.

Figure 1 shows the same pattern visually. Most points cluster around the diagonal, while one exact-zero actual posterior produces an extreme negative update and drives a large portion of the all-row RMSE.

5 Distribution-Conditional Diagnostics

The 200-run suggests a distribution-conditional story: the model predicts ordinary updates well, while rare extreme updates are less stable. We therefore condition on $|\Delta_{\text{act}}|$ bins.

Table 2 and Figure 2 are the key diagnostic evidence. There are 185 parseable pairs with $|\Delta_{\text{act}}| < 3$. In these ordinary regimes, Pearson correlations are 0.805 and 0.893 in the first two bins, with log-odds MAE below 0.47. The ≥ 5 bin contains only three examples and one exact-zero posterior; it is too small to support broad conclusions, but it explains why all-row Pearson and RMSE are more fragile than Spearman.

6 Worst-Case Examples

The worst cases are informative rather than merely embarrassing. The largest error is an exact 0.0 actual posterior, which becomes a very large negative log-odds shift after clipping. Other large errors often involve very small prior or posterior probabilities. This suggests that future runs should report both log-odds-space metrics and probability-space metrics, and should audit whether exact boundary outputs reflect genuine model certainty, parsing artifacts, or prompt-induced formatting behavior.

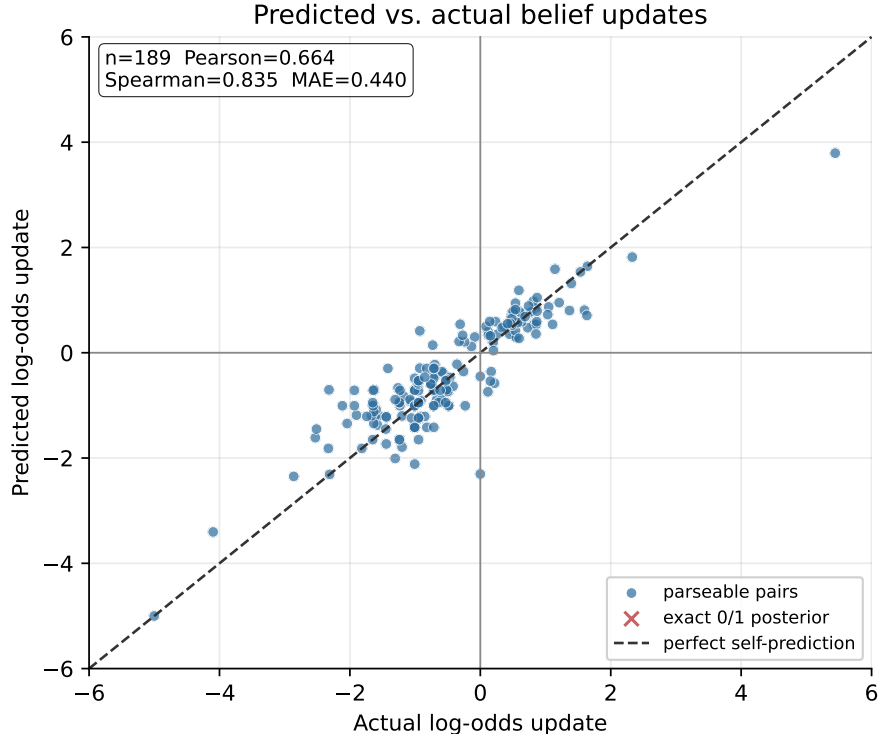


Figure 1: Predicted versus actual log-odds updates in the exact two-context protocol. The dashed line is perfect self-prediction. The exact 0/1 outlier is visible as a large negative actual shift.

Question	π	π'_{pred}	π'_{act}	Δ_{pred}	Δ_{act}
Scottie Scheffler wins 2026 Masters	0.220	0.340	0.000	0.602	-12.550
Belichick coaches Chargers in 2024-25	0.001	0.0001	0.001	-2.303	0.000
Solana reaches \$100 in March	0.300	0.950	0.990	3.792	5.442
Kim Jong Un out as Supreme Leader in 2025	0.020	0.010	0.002	-0.703	-2.321
New Coronavirus pandemic in 2025	0.020	0.030	0.008	0.416	-0.928

Table 3: Top worst-case examples by absolute log-odds shift error. Boundary and near-boundary probabilities create large log-odds effects even when probability differences are modest.

7 Related Work

LLM forecasting. Forecasting benchmarks evaluate whether LLMs can produce calibrated probabilities for future events. Halawi et al. show that retrieval-augmented LM systems can approach competitive human forecasting in some settings [1]. ForecastBench provides a dynamic benchmark of future-event forecasts designed to reduce contamination risk [2]. Our work is complementary: we do not primarily evaluate forecast accuracy, but whether a model can predict its own update when evidence is introduced.

Self-knowledge and self-prediction. Kadavath et al. study whether models can evaluate the correctness of their own answers and predict whether they know an answer [3]. More recent work asks whether models can predict their own behavior or learn about themselves by introspection [4, 5]. We adopt the self-prediction lens but apply it to probabilistic forecast revision rather than task success, answer validity, or output behavior.

Self-prediction degrades mainly in rare extreme updates

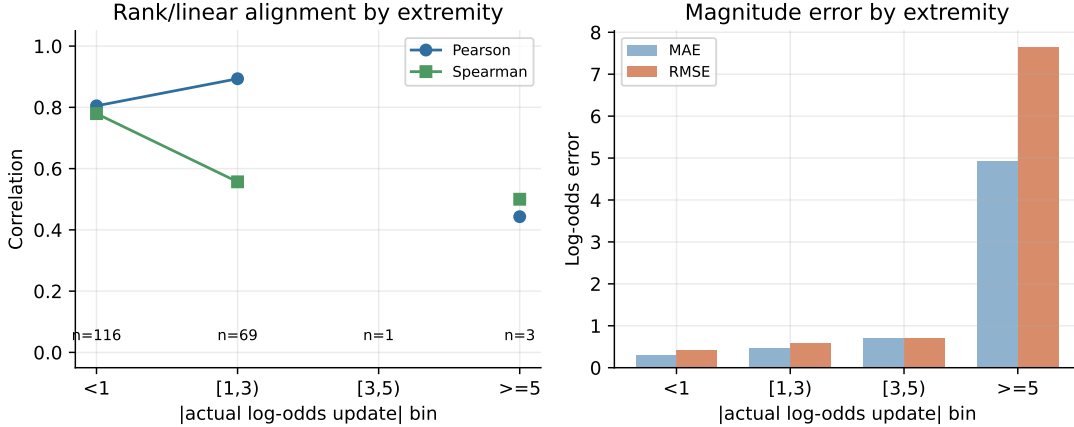


Figure 2: Correlations and error by actual-update extremity. The two-regime picture is visible: ordinary updates are predictable; rare extreme shifts are the main source of instability.

Bayesian and probabilistic reasoning in LLMs. Bayesian inference gives a normative account of belief revision under evidence. Recent work investigates how LLMs represent priors or learn probabilistic reasoning behavior [6, 7]. Our experiment is not a test of Bayes optimality. Instead, it asks whether the model’s own counterfactual prediction of its posterior update matches the posterior update that actually appears under the same evidence in a fresh context.

8 Limitations and Next Steps

This is a workshop-scale draft result with several limitations. First, we evaluate one model and neutral evidence only. The same protocol should be run on smaller open models and on weakly/strongly biased evidence framings. Second, the exact protocol uses generated evidence packets and a particular prompt format; prompt sensitivity remains an open question. Third, log-odds metrics are appropriate for forecast revision but highly sensitive near 0 and 1. We therefore report robust subsets and probability-space gaps, but future work should pre-register clipping rules and boundary handling. Finally, a model’s ability to predict its output under a prompt is not evidence of conscious self-knowledge. The claim is behavioral and operational: predicted update tracks realized update.

9 Conclusion

We introduced a two-context protocol for testing whether language models can forecast their own future belief updates. On 200 neutral ForecastBench-derived examples, GPT-OSS-120B predicts its own evidence-conditioned log-odds updates substantially better than simple baselines. Rank correlation remains high across the full run, and magnitude-sensitive degradation is concentrated in rare extreme shifts. These findings suggest a concrete and measurable form of metacognitive forecast revision: models can often say, before seeing evidence in an ordinary context, how that evidence will later move their own forecast.

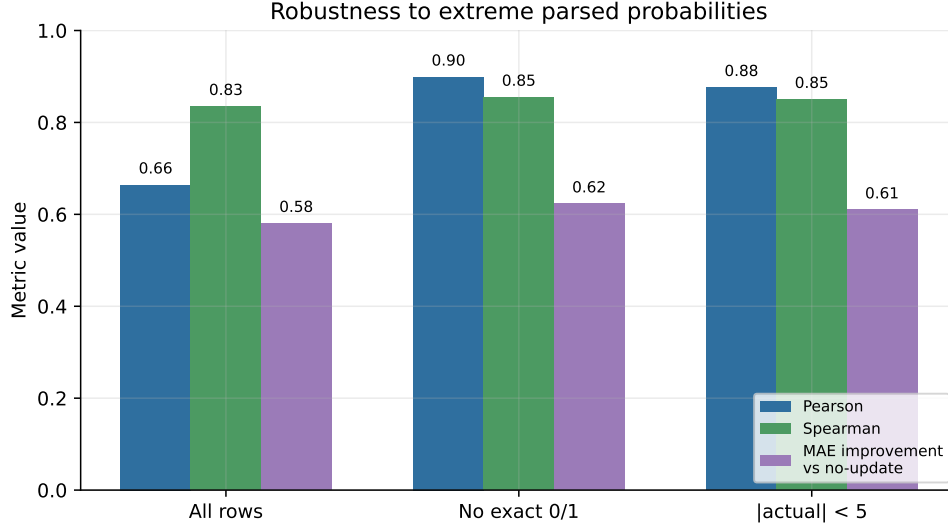


Figure 3: Robustness of the headline metrics. Excluding exact boundary posteriors or restricting to $|\Delta_{\text{act}}| < 5$ recovers high Pearson correlation while preserving high Spearman correlation and large improvement over no-update.

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